

112590024 邱紹澹 作業系統作業 HW4

10.44

```
OSHW_4_individual_112590024/10_44/alo.go

1 // go run alo.go <frame 數量>
2 package main
3
4 import (
5     "fmt"
6     "math/rand"
7     "os"
8     "strconv"
9 )

10_44 - zsh
kelen@KelendeiMac-Pro 10_44 % go run alo.go 3
Page Reference String (20 pages): 9, 0, 3, 3, 8, 4, 6, 3, 2, 8, 8, 0, 2, 3, 6, 2, 9, 5, 6, 9
Number of Frames: 3
-----
FIFO Page Faults: 16
LRU Page Faults: 15
OPT Page Faults: 11
kelen@KelendeiMac-Pro 10_44 %
```

10_44.png

$OPT \leq LRU \leq FIFO$

11.27

```
OSHW_4_individual_112590024/11_27/disk.go

1  // go run disk.go <初始磁頭位置>
2  package main
3  ↵
4  import (
5      "fmt"
6      "math/rand"
7      "os"
8      "sort"
9      "strconv"
10 )
11
12 const (
13     CYLINDERS    = 5000
14     NUM_REQUESTS = 1000
15 )
16

11_27 — zsh

kelen@KelendeiMac-Pro 10_44 % ls
10_44.png      alo.go
kelen@KelendeiMac-Pro 10_44 % cd ..
kelen@KelendeiMac-Pro OSHW_4_individual_112590024 % cd 11_27
kelen@KelendeiMac-Pro 11_27 % clear
kelen@KelendeiMac-Pro 11_27 % go run disk.go 2500
Cylinders:      0 to 4999
Total Requests: 1000
Initial Head:   2500
-----
FCFS  Total Head Movement: 1637463
SCAN  Total Head Movement: 7486
C-SCAN Total Head Movement: 9974
kelen@KelendeiMac-Pro 11_27 %
```

11_27

- **FCFS**：按照請求進來的順序一個一個服務，不做任何排序或優化
- **SCAN**：先對所有請求排序，從當前位置往高編號處理完，再回頭處理低編號的
- **C-SCAN**：同樣排序後往高編號處理，處理完後直接跳到最低編號，再繼續往高編號處理剩下的，不回頭